1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Coding:

```
import pandas as pd

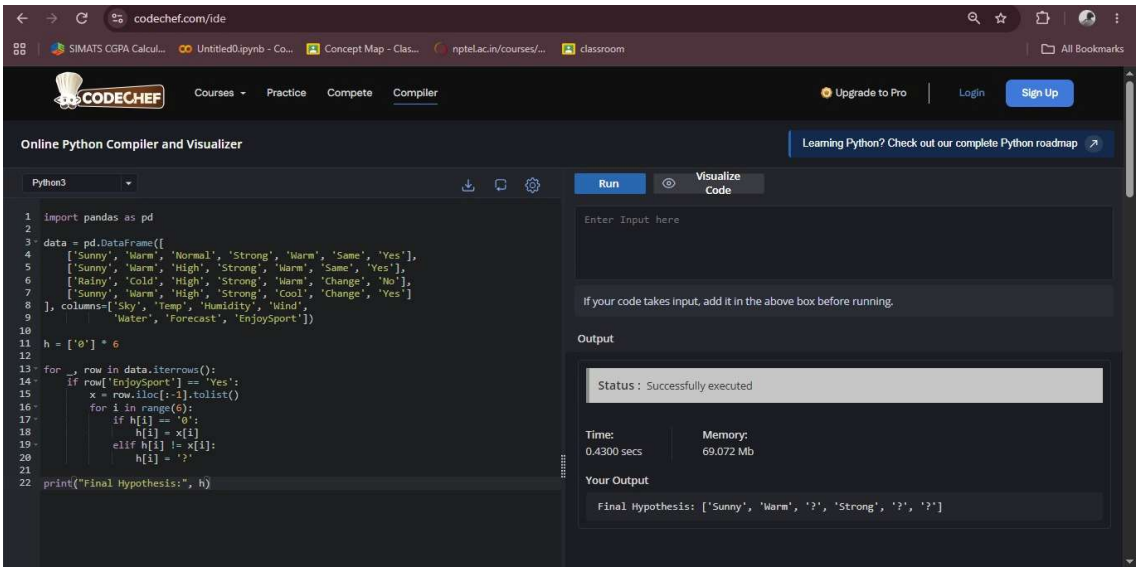
data = pd.DataFrame([
  ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
  ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
  ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
  ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
], columns=['Sky', 'Temp', 'Humidity', 'Wind',
       'Water', 'Forecast', 'EnjoySport'])

h = ['0'] * 6

for _, row in data.iterrows():
  if row['EnjoySport'] == 'Yes':
    x = row.iloc[:-1].tolist()
    for i in range(6):
      if h[i] == '0':
        h[i] = x[i]
      elif h[i] != x[i]:
        h[i] = '?'

print("Final Hypothesis:", h)
```

OUTPUT:




```python
import pandas as pd

data = pd.DataFrame([
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
], columns=['Sky', 'Temp', 'Humidity', 'Wind',
            'Water', 'Forecast', 'EnjoySport'])

h = ['0'] * 6

for _, row in data.iterrows():
    if row['EnjoySport'] == 'Yes':
        x = row.iloc[:-1].tolist()
        for i in range(6):
            if h[i] == '0':
                h[i] = x[i]
            elif h[i] != x[i]:
                h[i] = '?'

print("Final Hypothesis:", h)
```